

Deep Paresh Dodhiwala

213-561-7036 | deep.pd03007@gmail.com | [LinkedIn](#) | [Github](#) | [Portfolio](#)

EXPERIENCE

Software Engineer

Aug. 2024

Trimble Inc.

Princeton, NJ

- Designed and implemented **50+** full-stack features and bug fixes across two enterprise scheduling products (**Appian Resource Calendar**, **Appian Daily Planner**) — **Angular** UI, **C#/.NET** microservices, and **MongoDB** — backed by **AWS** updates and comprehensive test coverage on every change.
- Developed a scalable **E2E** testing framework for **ARC** and **ADP** using **Playwright (TypeScript)**, authoring **250+** automated tests over one year and cutting the automation backlog by **45%**.
- Monitored and stabilized the nightly **Bamboo** regression suite, triaging **100+** recurring failures to separate flaky tests from real defects, and authored **1000+** unit tests for **ARC** and **ADP**.

Algorithm Developer

Sept. 2023 – July. 2024

Autonomous Networks Research Group (ANRG)

Los Angeles, CA

- Researched and implemented task graph scheduling algorithms (**Min-Min and Sufferage**) for distributed computing environment within the SAGA (Scheduling Algorithms Gathered) project.
- Extended the scheduling algorithms with financial cost constraints (**E-HEFT**), improving task execution efficiency and load balancing over baseline heuristics

Software Engineer Intern

May 2021 – July 2021

JPMorgan Chase

Mumbai, India

- Enhanced **Financial Asset Record Software** by analyzing and recommending architectural modifications to incorporate Sub-Line-Of-Business as an independent component during the End-Of-Day functionality.
- Proposed and designed critical code changes in **Spring Boot** and database structure modifications in **Apache Cassandra** to ensure accurate and efficient operations.

TECHNICAL SKILLS

Languages: Python, C#, Java, TypeScript, C, C++, JavaScript, Bash, PHP.

Frameworks & Libraries: ASP.NET, Angular, Playwright, Node.js, React.js, Flask, TensorFlow, PyTorch.

Databases: MongoDB, MySQL, Oracle, PostgreSQL, Cassandra, Amazon DynamoDB.

Tools & Platforms: Git, Jira, Bamboo, AWS, Docker, Jenkins, Jest.

PROJECTS

Weenix Operating System | C, Git

- Implemented core Linux-based Operating System functions using kernel programming in C such as **Processes**, **Multithreading**, **Virtual File System** and **Virtual Memory**.
- The OS cleared all the stress tests like Mem-eating kernel and fork-bombing 500 processes.

Intelligent Call Prioritization Using Speech Emotion Recognition | Python, Anvil, Git

- Developed a website using **Anvil** that uses a short voice note of waiting callers and prioritizes calls based on emotion predicted through speech and transcribed text.
- Deployed BERT model for **textual analysis** and Random Forest Classifier for the **Speech Emotion Recognition** for the best results.
- The combined model achieved an overall accuracy of **92%** and with average waiting time of **30 seconds**, a **1.5x** improvement on the standard scheduling algorithms.

Eggshub | Angular, Node.js, MongoDB, Plotly

- Led a team of six to develop a data-driven platform providing comprehensive insights into global egg trade for a non-profit organization (Code for Good Hackathon).
- Designed and implemented backend APIs to serve global egg trade data and developed interactive choropleth maps using **Plotly**, enhancing data accessibility and interpretation.

EDUCATION

University of Southern California (USC)

Los Angeles, CA

Master of Science in Computer Science, CGPA: 3.75/4

Aug. 2022 – May 2024

Courserwork: Analysis of Algorithms, Operating Systems, Machine Learning, Database Systems.

Sardar Patel Institute of Technology

Mumbai, India

Bachelor of Technology in Information Technology, CGPA: 9.5 / 10

Aug. 2018 – May 2022